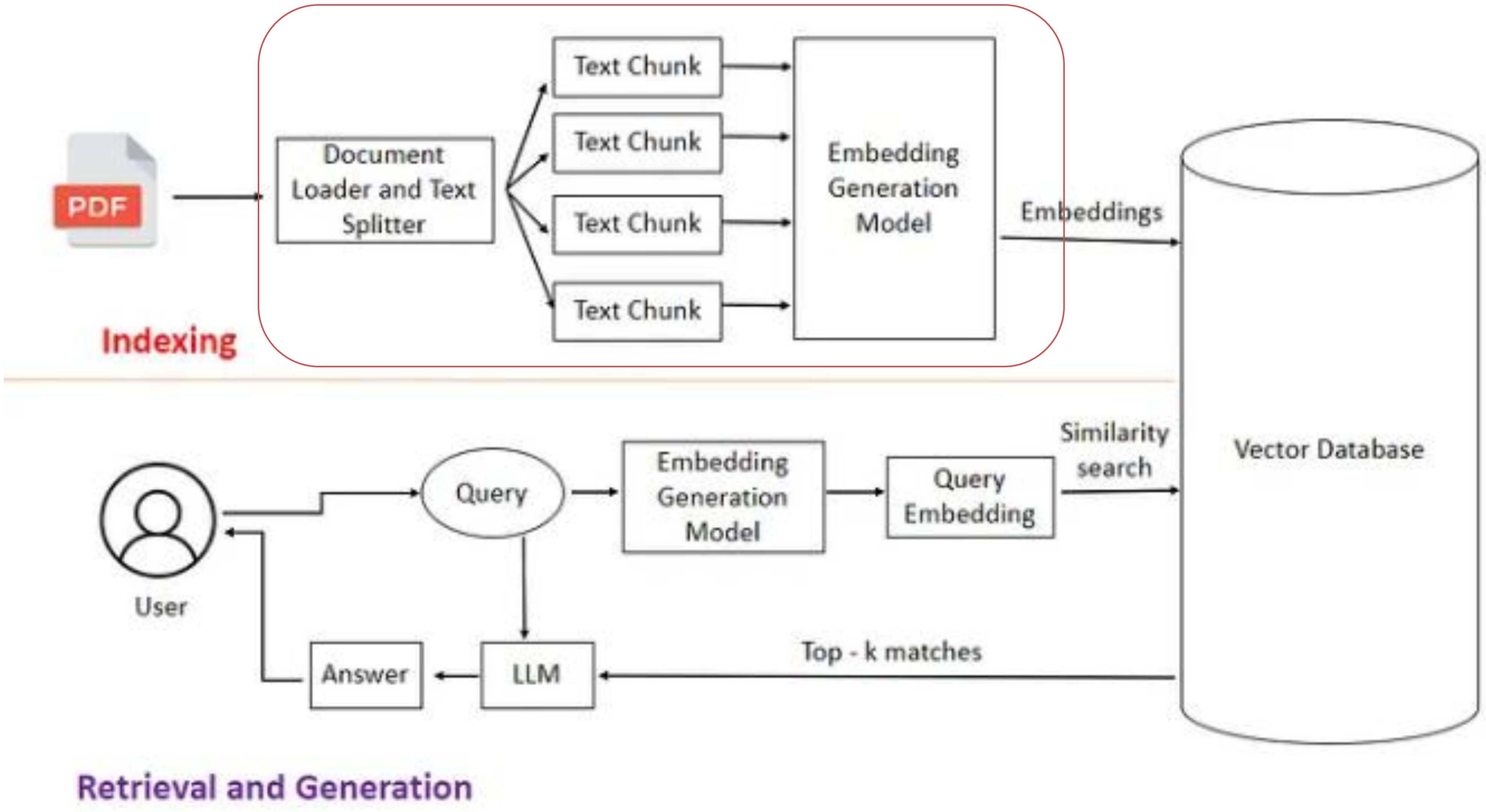
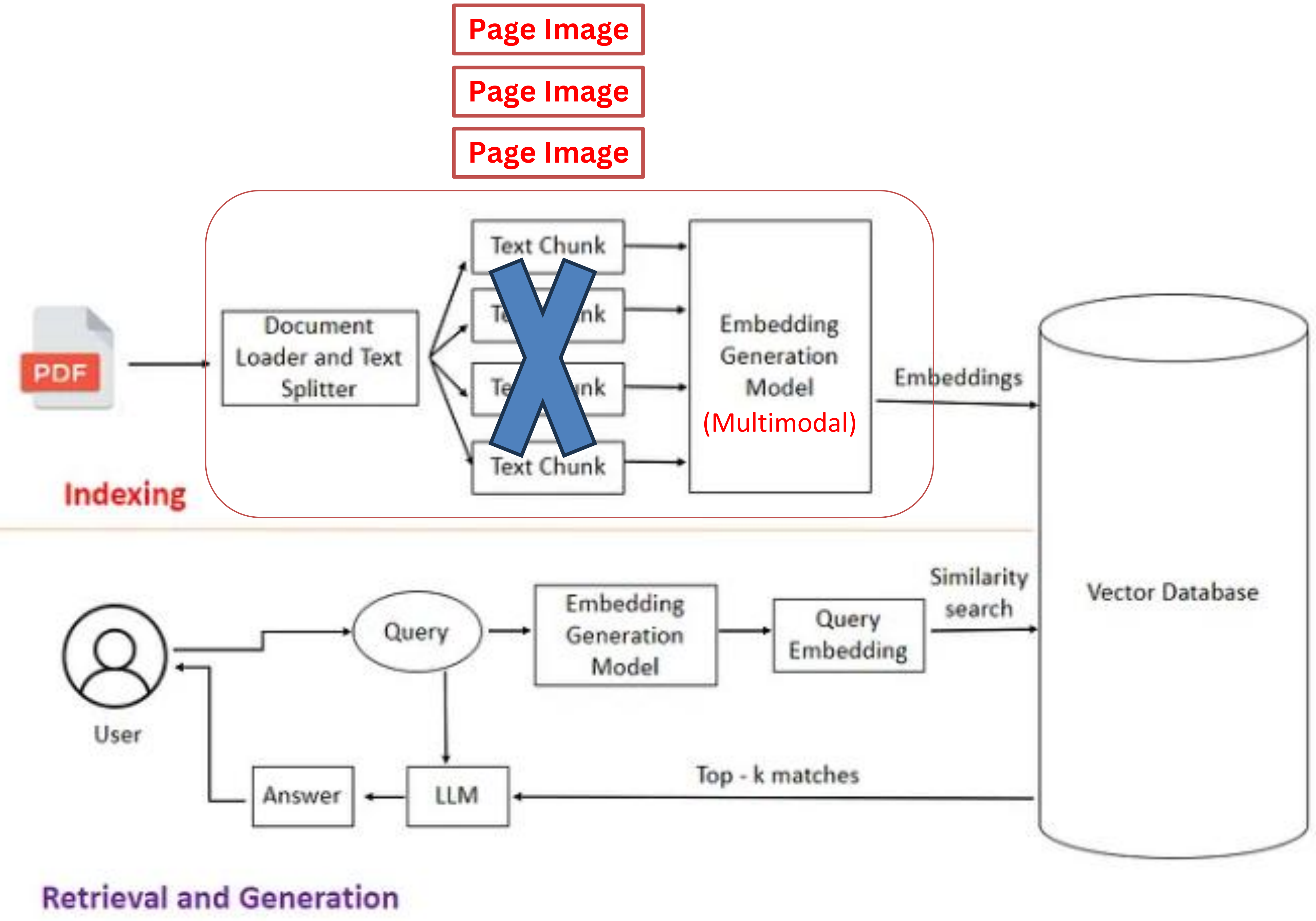
Visual-RAG in Enterprise Domain LLM



The Existing LLM used a **text-based RAG**: Store to DB by OCR-based text extraction

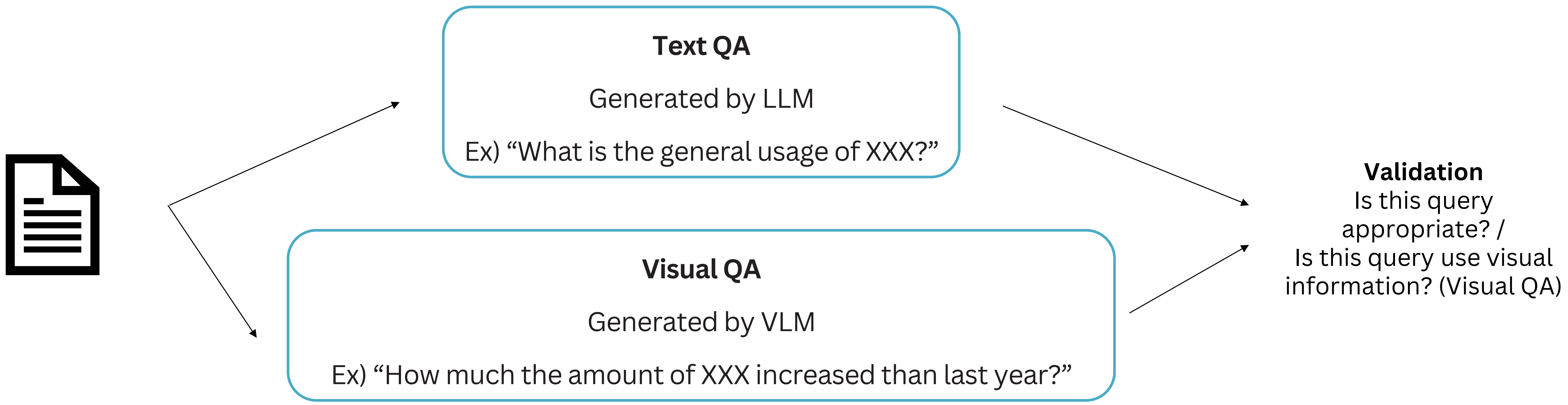
However, most documents contain important **visual information (**tables, charts, diagrams...)

→ Can we retrieve **document page image directly**?



Preserves visual information &

Reduces preprocessing cost by skipping OCR.



Need Korean benchmark, Verify both visual/text performance.

Randomly sampled about **8,000 internal document pages** and generated **1,000 QA pairs** by VLM.

(600 Text QA, 400 Visual QA)

Metrics: **Recall@1, 5, 10**

Result

Text QA (600 Query)

	recall@1	recall@5	recall@10
Qwen3-4B (Text)	0.50 (297)	0.73 (438)	0.79 (390)
Jina-embedder-v4-4B (Vision)	0.62 (373)	0.87 (521)	0.92 (551)
Qwen3-VL-Embedding-8B (Vision)	**0.66 (396)**	**0.91 (545)**	**0.96 (573)**

Visual QA (400 Query)

	recall@1	recall@5	recall@10
Qwen3-4B (Text)	0.28 (111)	0.47 (189)	0.55 (218)
Jina-embedder-v4-4B (Vision)	**0.49 (197)**	0.75 (301)	0.82 (329)
Qwen3-VL-Embedding-8B (Vision)	0.49 (194)	**0.76 (302)**	**0.83 (330)**

Vision-RAG retrieves document pages more accurately,

especially when the query depends **on visual information.**